

cloud. ownCloud has a Web-based front end that is built on top of PHP, and an SQLite, MySQL, Oracle or PostgreSQL database. What this means is that you can set up the ownCloud server on any platform that can support these technologies. The server side installer is also fairly simple and doesn't really require any advanced technical knowhow. Besides the regular syncing features, it also allows you to install custom apps that can extend its functionality by adding features like task lists, etc. A full demo of the ownCloud Web console can be seen on the website.

Setting up a Web server

As mentioned earlier, ownCloud is a Web-based system and, hence, the simplest way to set it up is on a LAMP (Linux, Apache2, MySQL, PHP/Perl/Python) stack. This popular acronym refers to a package that has all the principal ingredients; the Apache2 Web server, the MySQL database and the PHP support that are needed on a Linux-based server machine, turning it into a generic, free and open source Web server. The concept of a LAMP stack has now been made available on the Mac and Windows OSs as well.

Linux (Ubuntu)

Open up the Ubuntu dash (the button with the Ubuntu logo on the left hand panel) and search for 'Terminal'. Fire up the system terminal.

Step 1: Before installing the LAMP stack or any other software on Ubuntu, it is always prudent to perform a few system updates and upgrades. In the window that opens up, type the following commands:

```
sudo apt-get update
sudo apt-get upgrade
```

apt/Advanced Packaging Tools is the name of the package that contains a set of tools used by Ubuntu (along with other variants of Debian Linux) to handle software installation and removal. The command update re-syncs the package files with its sources (links to software repositories). The upgrade process then continues, installing any new package files that are discovered during the sync operation performed earlier.

Step 2: Run the following command to install the LAMP stack:

```
sudo apt-get install lamp-server^
```

Please note the ^ at the end of the package name. While the set-up runs, it will ask you for the MySQL root password. As with any password, enter something that is secure as well as easy to recall for you, but hard for others to guess.

Step 3: Now that the LAMP stack is installed, prepare a MySQL database for ownCloud. As mentioned earlier, ownCloud supports SQLite, MySQL, PostgreSQL and Oracle, but for the purposes of this article, we will use MySQL as it is included as a part of the LAMP stack and is faster than

SQLite. You can choose to use any of the other supported databases mentioned as well.

Run the following command at the terminal:

```
sudo mysql_secure_installation
```

This should prompt you for the MySQL root password you had created in Step 2. Enter the password and type 'n' when you are asked to change the root password. Type 'y' for all the other responses.

Step 4: The final step before getting ownCloud up and running is installing the required libraries on the server:

```
sudo apt-get install php5-gd php-xml-parser php5-intl
sudo apt-get install curl libcurl3 php5-curl
```

The next thing to do is to enable *mod_rewrite* and *mod_headers* on the Apache2 server. These modules will enable ownCloud to use Apache's *.htaccess* files, which it needs to function properly. A *.htaccess* file is used by Apache to allow websites to configure their access details without needing to change/access the *server config* files. They are very useful in a shared server set-up where, typically, people don't have access to the *server config*.

Type the following commands in the terminal to enable *mod_rewrite* and *mod_headers*:

```
sudo a2enmod rewrite
sudo a2enmod headers
sudo nano /etc/apache2/sites-available/default
```

On running the last command, Ubuntu's in-terminal text editor should open up the Apache2 *config* file. Find the line `<Directory /var/www/>` and change *AllowOverride* to *All* from *None*. Save the file by pressing *Ctrl+X*, followed by *Ctrl+Y* and then *Enter*.

Restart Apache2 by entering the following command:

```
sudo service apache2 restart
```

Mac/Windows

The process of installing a Web server is fairly simple on a Mac/Windows system, since the LAMP stack has been pre-packaged as an installer file for these systems and appropriately renamed as WAMP (Windows) and MAMP (Mac OSX). The links to download these applications can be found at the end of the article.

Install the packaged installer containing PHP 5.4 and Apache 2.4 and run the application installed to start the server.

Set up the ownCloud database

Linux (Ubuntu)

In a terminal, type the following command to log in to MySQL: